
Cyclic Discrete-Goldstone Protection of QCD Chronometric Shear

A Z_6 replicated-sector model preserving the 2/27 threshold while suppressing the ratio-mode mass

Main conclusion. The radiative obstruction isolated in v0.5 can be removed by replacing the unprotected ratio field with a compact scalar that nonlinearly realizes an exact cyclic exchange symmetry. The complete N -sector orbit cancels every field-dependent Coleman-Weinberg term below order ϵ^N , while only one vectorlike fundamental is charged under visible QCD. Consequently,

$$V_{\text{vac}}(a) = O(\epsilon^N M^4), \quad \frac{\partial}{\partial a} \ln \frac{\Lambda_{3,0}}{\chi} = \left[\frac{2}{27} + O(\alpha_s) \right] \frac{\partial}{\partial a} \ln \frac{M_0}{\chi}.$$

For the minimal clean benchmark $N = 6$ at $a_0/f_a = \pi/2$,

$$m_a^2 = \frac{27}{320\pi^2} \frac{M^4}{f_a^2} \epsilon^6, \quad d_g = \frac{2}{27} \frac{M_{\text{P}}}{f_a} \epsilon.$$

The dangerous $O(\epsilon^2 M^4/f_a^2)$ mass is absent by symmetry rather than counterterm tuning. The remaining hard problems are ultraviolet symmetry quality, replicated-sector cosmology and finite-density screening.

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Core construction:	Exact Z_N exchange symmetry, N replicated colour sectors, one vectorlike triplet per sector, a compact ratio field, and standard QCD threshold matching
Main calculable result:	Visible response remains $2/27 + O(\alpha_s)$ while the vacuum mass begins at ϵ^N
Central limitation:	The elementary compact-scalar version has a quality problem; a discrete-gauge or deconstructed Wilson-line completion is required

Abstract

We construct a conventional local $3 + 1$ -dimensional protection sector for the chronometric QCD ratio mode. The theory contains N identical colour or Standard-Model sectors, cyclically exchanged by an exact Z_N , together with one vectorlike Dirac fundamental Ψ_k per sector and a compact scalar a/f_a . Their masses form a complete orbit,

$$M_k = M [1 - \epsilon \cos(a/f_a + 2\pi k/N)].$$

The full vacuum functional sums over the orbit, whereas visible experiments probe only sector zero. Root-of-unity identities and an all-loop spurion argument imply that the first nonconstant vacuum harmonic is proportional to $\epsilon^N \cos(Na/f_a)$. At one loop and for $N > 4$,

$$V_{\text{CW}} = \frac{N_c M^4 \epsilon^N}{8\pi^2} F_N \cos(Na/f_a) + O(\epsilon^{N+1}), \quad F_N = \frac{24 \cdot 2^{1-N}}{(N-1)(N-2)(N-3)(N-4)}.$$

The hidden partners are charged under separate colour groups, so they do not alter the visible beta function. The exact threshold recursion of v0.5 therefore remains valid and gives $d \ln(\Lambda_{3,0}/\chi) = [2/27 + O(\alpha_s)] d \ln(M_0/\chi)$. For $N = 6$, the symmetry-related vacuum $a_0/f_a = \pi/2$ gives an exactly maximal linear threshold slope, $\partial \ln M_0 / \partial(a/f_a) = \epsilon$, while $m_a^2 = (27/320\pi^2) M^4 \epsilon^6 / f_a^2$. The ordinary one-threshold mass is suppressed by $(9/80)\epsilon^4$ at the squared-mass level. All symbolic identities, Fourier coefficients, mass positivity and numerical benchmarks were independently checked. The mechanism itself has close literature precedents; the candidate contribution is its CP-even QCD-threshold realization, retention of the $2/27$ transmission coefficient, and embedding in the clock-equivalence-principle framework.

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1 Executive result

The radiative-naturalness obstruction in v0.5 can be removed without sacrificing the calculable visible-QCD transmission coefficient.

Replace the unprotected real ratio mode by a compact field

$$x \equiv \frac{a}{f_a}$$

that nonlinearly realizes an exact cyclic symmetry Z_N . Introduce N identical QCD-like sectors and one vectorlike colour triplet Ψ_k in each sector, with masses

$$M_k(a) = M \left[1 - \epsilon \cos \left(x + \frac{2\pi k}{N} \right) \right], \quad k = 0, \dots, N-1.$$

The cyclic transformation is

$$x \rightarrow x + \frac{2\pi}{N}, \quad k \rightarrow k + 1.$$

The full vacuum energy sums over the complete orbit. Root-of-unity identities cancel every field-dependent Coleman-Weinberg term below order ϵ^N . For $N > 4$,

$$V_{\text{CW}}(a) = \frac{N_c M^4 \epsilon^N}{8\pi^2} F_N \cos \left(\frac{Na}{f_a} \right) [1 + O(\epsilon)],$$

where

$$F_N = \frac{24 \cdot 2^{1-N}}{(N-1)(N-2)(N-3)(N-4)}.$$

Therefore,

$$m_a^2 = \frac{N_c M^4 N^2 \epsilon^N}{8\pi^2 f_a^2} F_N [1 + O(\epsilon)].$$

The coupling of a to **our** QCD sector is not orbit-summed. Only Ψ_0 is charged under visible $SU(3)_0$; its partners are charged under the hidden groups $SU(3)_k$. Consequently the visible threshold remains

$$\text{d ln } \frac{\Lambda_{3,0}}{\chi} = \left[\frac{2}{27} + O(\alpha_s) \right] \text{d ln } \frac{M_0}{\chi}.$$

Thus the scalar coupling is order ϵ , while its radiative mass is order $\epsilon^{N/2}$:

$$\text{visible QCD response} \sim \epsilon, \quad m_a^2 \sim \epsilon^N.$$

For the minimal especially clean choice $N = 6$, select the symmetry-related vacuum

$$\frac{a_0}{f_a} = \frac{\pi}{2}.$$

Then

$$\frac{\partial}{\partial(a/f_a)} \ln \frac{M_0}{\chi} = \epsilon,$$

while

$$m_a^2 = \frac{27}{320\pi^2} \frac{M^4}{f_a^2} \epsilon^6 [1 + O(\epsilon)]$$

for $N_c = 3$. The visible gluonic response is

$$d_g \equiv M_P \frac{\partial}{\partial a} \ln \frac{\Lambda_{3,0}}{\chi} = \frac{2}{27} \frac{M_P}{f_a} \epsilon + O(\alpha_s \epsilon, \epsilon^2).$$

The catastrophic one-threshold estimate

$$m_{a,\text{naive}}^2 \sim \frac{N_c}{4\pi^2} \frac{M^4}{f_a^2} \epsilon^2$$

is replaced, in the Z_6 model, by

$$\frac{m_{a,Z_6}^2}{m_{a,\text{naive}}^2} = \frac{9}{80} \epsilon^4$$

under the stated normalization. This is a symmetry cancellation, not a tuning between unrelated counterterms.

2 Status matrix

Requirement	Verdict	Result
Preserve visible-QCD 2/27 threshold	PASS	Hidden partners live in separate colour sectors, so only one fundamental Dirac threshold enters visible QCD.
Cancel M_Ψ^4/f_a^2 mass problem	PASS parametrically	All terms through $O(\epsilon^{N-1})$ cancel; the first vacuum harmonic is $O(\epsilon^N)$.
Conventional Lorentzian unitarity	PASS	Two-derivative scalar, gauge and vectorlike-fermion sectors; no higher-time-derivative elementary field.

Requirement	Verdict	Result
Positive physical spectrum	PASS perturbatively	Canonical kinetic terms, anomaly-free vectorlike representations, positive radial curvature and positive angular curvature at a selected minimum.
All-loop protection	PASS conditionally	Follows from exact Z_N , the restored continuous shift at $\epsilon = 0$, and absence of independent shift-breaking spurions.
Renormalizable protection sector	PASS	The displayed scalar, Yukawa and gauge interactions are dimension four.
UV quality against Φ^N operators	PARTIAL	An elementary global-field version has a quality problem; a discrete-gauge or deconstructed Wilson-line completion is required.
Mirror-sector cosmology	OPEN	Reheating, dark radiation, hidden confinement and domain selection need a dedicated model.
Finite-density behavior	OPEN but calculable	Visible matter spontaneously selects one sector and gives an unsuppressed environmental potential; this is both the signal source and a possible screening mechanism.
Novelty	NARROW	The Z_N protection mechanism is established. The CP-even QCD-threshold implementation, preserved 2/27 transmission and chronometric/EP application appear to be the candidate new package.

3 Why the single-threshold model failed

The v0.5 model used one vectorlike colour triplet with

$$M_\Psi(a) = M \left[1 + \varepsilon_\Psi \frac{a}{f_a} + \cdots \right].$$

Its one-loop vacuum energy contains

$$V_{\text{CW}} = -\frac{N_c}{16\pi^2} M_\Psi(a)^4 \left[\ln \frac{M_\Psi(a)^2}{\mu^2} - \frac{3}{2} \right].$$

Taking two derivatives gives generically

$$m_a^2 \sim \frac{N_c}{4\pi^2} \varepsilon_\Psi^2 \frac{M^4}{f_a^2}.$$

The same derivative that produces the useful QCD coupling therefore produces a large scalar mass. With a TeV-scale or higher coloured threshold, an astronomical-range field demands absurdly small ε_Ψ , and the observable signal vanishes with it.

The required structure must therefore separate:

first derivative of one visible threshold

from

second derivative of the complete vacuum energy.

A cyclic exchange symmetry does exactly this. The visible experiment samples one sector; the vacuum sums the complete symmetry orbit.

4 Microscopic 3 + 1-dimensional construction

4.1 Gauge and matter content

Take

$$\mathcal{G} = \left[\prod_{k=0}^{N-1} SU(3)_k \right] \rtimes Z_N.$$

For exact radiative protection, every sector must contain an identical renormalization environment for the protected heavy fermion. The conservative completion replicates the complete coloured sector, and the simplest known cosmological implementations replicate the full Standard Model:

$$\mathcal{L}_{\text{sectors}} = \sum_{k=0}^{N-1} \left[\mathcal{L}_{\text{SM}}^{(k)} + \bar{\Psi}_k i \not{D}_k \Psi_k \right].$$

Each Ψ_k is a vectorlike Dirac fundamental of $SU(3)_k$. It can be assigned electroweak quantum numbers permitting decay, provided those assignments and decay operators are replicated identically.

Introduce:

- the universal scale field χ ;
- a complex scalar Φ , neutral under the colour sectors;
- a radial locking potential

$$V_{\text{rad}} = \lambda_\Phi \left(|\Phi|^2 - \frac{\zeta_a^2 \chi^2}{2} \right)^2, \quad \lambda_\Phi > 0.$$

Write

$$\Phi = \frac{f_a + \rho}{\sqrt{2}} \exp\left(\frac{ia}{f_a}\right), \quad f_a = \zeta_a \chi$$

in the adiabatic locked background.

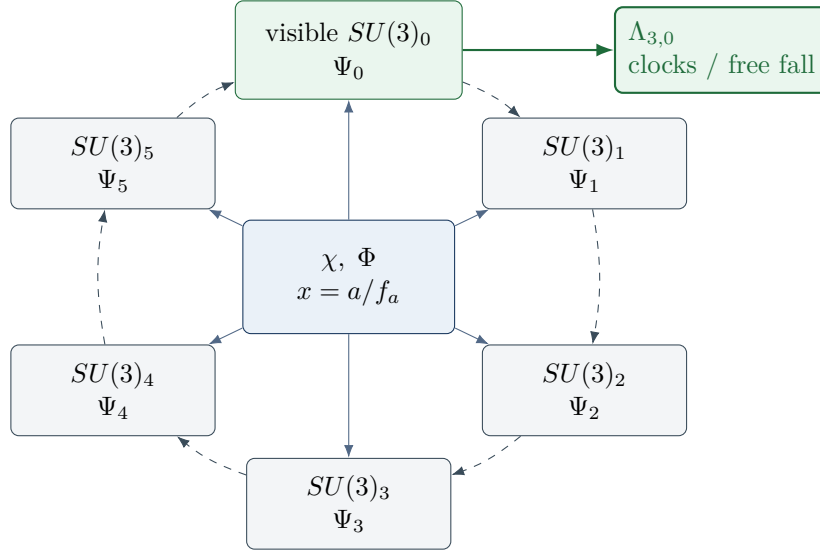


Figure 1: The complete Z_6 orbit protects the vacuum potential. Only the highlighted sector contributes to the visible-QCD threshold response.

4.2 Exact cyclic symmetry

Let

$$\omega = e^{2\pi i/N}.$$

The symmetry acts as

$$\boxed{\Phi \rightarrow \omega \Phi, \quad \Psi_k \rightarrow \Psi_{k+1}, \quad SU(3)_k \rightarrow SU(3)_{k+1}.}$$

The renormalizable Yukawa orbit is

$$\boxed{\mathcal{L}_Y = - \sum_{k=0}^{N-1} \left[y_\chi \chi - y \left(\omega^k \Phi + \omega^{-k} \Phi^\dagger \right) \right] \bar{\Psi}_k \Psi_k.}$$

After radial locking,

$$M = y_\chi \chi, \quad \epsilon = \frac{\sqrt{2} y f_a}{M} = \frac{\sqrt{2} y \zeta_a}{y_\chi},$$

and

$$M_k(a) = M \left[1 - \epsilon \cos \left(\frac{a}{f_a} + \frac{2\pi k}{N} \right) \right].$$

All masses remain strictly positive for

$$0 < \epsilon < 1.$$

Because $M \propto \chi$ and $f_a \propto \chi$, ϵ is dimensionless and independent of the common radial scale. The compact field a/f_a is therefore a genuine ratio mode.

4.3 Perturbative health

The scalar kinetic term gives

$$|\partial\Phi|^2 = \frac{1}{2}(\partial\rho)^2 + \frac{1}{2}\left(1 + \frac{\rho}{f_a}\right)^2 (\partial a)^2.$$

Around $\rho = 0$, both kinetic eigenvalues are positive. The radial mass is

$$m_\rho^2 = 2\lambda_\Phi f_a^2 > 0.$$

The Ψ_k are vectorlike, so their colour representations introduce no chiral gauge anomaly. Each gauge sector is an ordinary local Yang-Mills theory. The construction therefore avoids the Lorentzian and spectral difficulties associated with elementary fourth-order fields.

5 All-orders cyclic-protection theorem

5.1 Theorem

Assume:

1. the complete microscopic action, path-integral measure and regulator preserve the cyclic Z_N ;
2. the limit $\epsilon \rightarrow 0$ restores a continuous shift symmetry of a ;
3. the only local shift-breaking spurions are those appearing in the complete Yukawa orbit;
4. the effective action is analytic in those spurions near $\epsilon = 0$.

Then the vacuum effective potential has the form

$$V_{\text{eff}}(a) = V_0 + \sum_{q=1}^{\infty} \Lambda_q^4 \cos \left(\frac{qNa}{f_a} + \delta_q \right),$$

with

$$\Lambda_q^4 = O(\epsilon^{qN}).$$

In particular, the leading field-dependent term is $O(\epsilon^N)$.

5.2 Proof

The exact exchange symmetry implies

$$V_{\text{eff}}(x) = V_{\text{eff}}\left(x + \frac{2\pi}{N}\right).$$

Its Fourier series can therefore contain only harmonics e^{iqNx} . In the $\epsilon = 0$ limit, the continuous shift symmetry makes every nonconstant Fourier coefficient vanish.

Each Yukawa mass insertion supplies one factor of ϵe^{ix} or ϵe^{-ix} . A term proportional to e^{iqNx} requires at least qN net insertions. Hence its coefficient is at least order ϵ^{qN} . This argument is independent of loop order. Gauge and scalar corrections can alter coefficients but cannot create a forbidden harmonic or reduce the required spurion power.

5.3 Consequence

The dangerous order- ϵ^2 mass is not balanced against an unrelated bosonic loop. It is absent because no symmetry-compatible invariant exists at that order.

6 Explicit one-loop Coleman-Weinberg potential

For N_c colours,

$$V_{\text{CW}}(a) = -\frac{N_c}{16\pi^2} \sum_{k=0}^{N-1} M_k(a)^4 \left[\ln \frac{M_k(a)^2}{\mu^2} - \frac{3}{2} \right].$$

For $m < N$, the orbit sums obey

$$\sum_{k=0}^{N-1} \cos^m\left(x + \frac{2\pi k}{N}\right) = \text{constant in } x.$$

The first field-dependent identity is

$$\sum_{k=0}^{N-1} \cos^N\left(x + \frac{2\pi k}{N}\right) = \frac{N}{2^{N-1}} \cos(Nx) + \text{constant}.$$

For $N > 4$, the divergent and renormalization-scale-dependent terms are field independent, and expansion of the finite part gives

$$V_{\text{CW}}(a) = \frac{N_c M^4 \epsilon^N}{8\pi^2} F_N \cos(Nx) [1 + O(\epsilon)].$$

The literature expression

$$F_N = 2^{1-N} N \sum_{\ell=0}^4 \binom{4}{\ell} \frac{(-1)^\ell}{N - \ell}$$

simplifies exactly to

$$F_N = \frac{24 \cdot 2^{1-N}}{(N-1)(N-2)(N-3)(N-4)}.$$

At a minimum,

$$m_a^2 = \frac{N_c M^4 N^2 \epsilon^N}{8\pi^2 f_a^2} F_N [1 + O(\epsilon)] > 0.$$

The symmetry statement is all-order; this coefficient is the leading one-loop realization.

7 Why the visible 2/27 survives

The partners $\Psi_{k \neq 0}$ cancel the vacuum potential because the vacuum functional sums over all sectors. They do **not** enter the visible QCD beta function because they are not charged under $SU(3)_0$.

For our sector, the exact threshold recursion from v0.5 remains

$$\delta_{n-1} = A_j \delta_n + (1 - A_j) \epsilon_j,$$

where

$$\delta_n = d \ln \frac{\Lambda_n}{\chi}, \quad \epsilon_j = d \ln \frac{\hat{m}_j}{\chi}.$$

For the visible threshold chain

$$7 \xrightarrow{\Psi_0} 6 \xrightarrow{t} 5 \xrightarrow{b} 4 \xrightarrow{c} 3,$$

with the Standard Model heavy quarks exactly locked to χ ,

$$d \ln \frac{\Lambda_{3,0}}{\chi} = \mathcal{D}_{\Psi \rightarrow 3} d \ln \frac{M_0}{\chi}.$$

At one loop,

$$\begin{aligned} \mathcal{D}_{\Psi \rightarrow 3}^{(1)} &= \left(1 - \frac{19}{21}\right) \frac{21}{23} \frac{23}{25} \frac{25}{27} \\ &= \boxed{\frac{2}{27}}. \end{aligned}$$

At higher orders,

$$\mathcal{D}_{\Psi \rightarrow 3} = \frac{2}{27} + O(\alpha_s(M)),$$

with corrections calculable from standard decoupling functions.

For a general background $x_0 = a_0/f_a$, define

$$\kappa_0 \equiv \left. \frac{\partial}{\partial x} \ln \frac{M_0}{\chi} \right|_{x_0} = \frac{\epsilon \sin x_0}{1 - \epsilon \cos x_0}.$$

Then

$$\boxed{\frac{\partial}{\partial a} \ln \frac{\Lambda_{3,0}}{\chi} = \left[\frac{2}{27} + O(\alpha_s) \right] \frac{\kappa_0}{f_a}.$$

This is the key decoupling:

$$\kappa_0 = O(\epsilon), \quad m_a^2 = O(\epsilon^N).$$

8 Minimal working benchmark: Z_6

For $N = 6$,

$$F_6 = \frac{1}{160}.$$

The potential has six minima

$$x_m = \frac{(2m-1)\pi}{6}.$$

Choose the symmetry-related vacuum

$$\boxed{x_0 = \frac{\pi}{2}}.$$

Then

$$\cos x_0 = 0, \quad \sin x_0 = 1,$$

so

$$M_0 = M, \quad \boxed{\kappa_0 = \epsilon}.$$

For $N_c = 3$,

$$\boxed{V_{\text{CW}}^{(6)} = \frac{3M^4\epsilon^6}{1280\pi^2} \cos\left(\frac{6a}{f_a}\right) + O(\epsilon^7),}$$

and

$$\boxed{m_a^2 = \frac{27}{320\pi^2} \frac{M^4}{f_a^2} \epsilon^6 + O(\epsilon^7).}$$

The QCD coupling is

$$d_g = \frac{2}{27} \frac{M_P}{f_a} \epsilon + O(\alpha_s \epsilon, \epsilon^2).$$

Eliminating ϵ gives the model's mass-coupling protection law:

$$\frac{m_a^2 f_a^2}{M^4} = \frac{27}{320\pi^2} \left(\frac{27}{2} \frac{f_a}{M_P} d_g \right)^6 + \dots$$

This is the central quantitative gain: the coupling enters the radiative mass at sixth order rather than second order.

8.1 Numerical illustration

Take

$$M = 10 \text{ TeV}, \quad f_a = M_P = 2.435 \times 10^{18} \text{ GeV}.$$

N	ϵ	d_g at maximal slope	m_a	Compton range
5	10^{-6}	7.41×10^{-8}	$1.00 \times 10^{-17} \text{ eV}$	0.132 AU
6	10^{-6}	7.41×10^{-8}	$3.80 \times 10^{-21} \text{ eV}$	347 AU
8	10^{-6}	7.41×10^{-8}	$9.57 \times 10^{-28} \text{ eV}$	$1.38 \times 10^9 \text{ AU}$
6	10^{-3}	7.41×10^{-5}	$3.80 \times 10^{-12} \text{ eV}$	$3.47 \times 10^{-7} \text{ AU}$

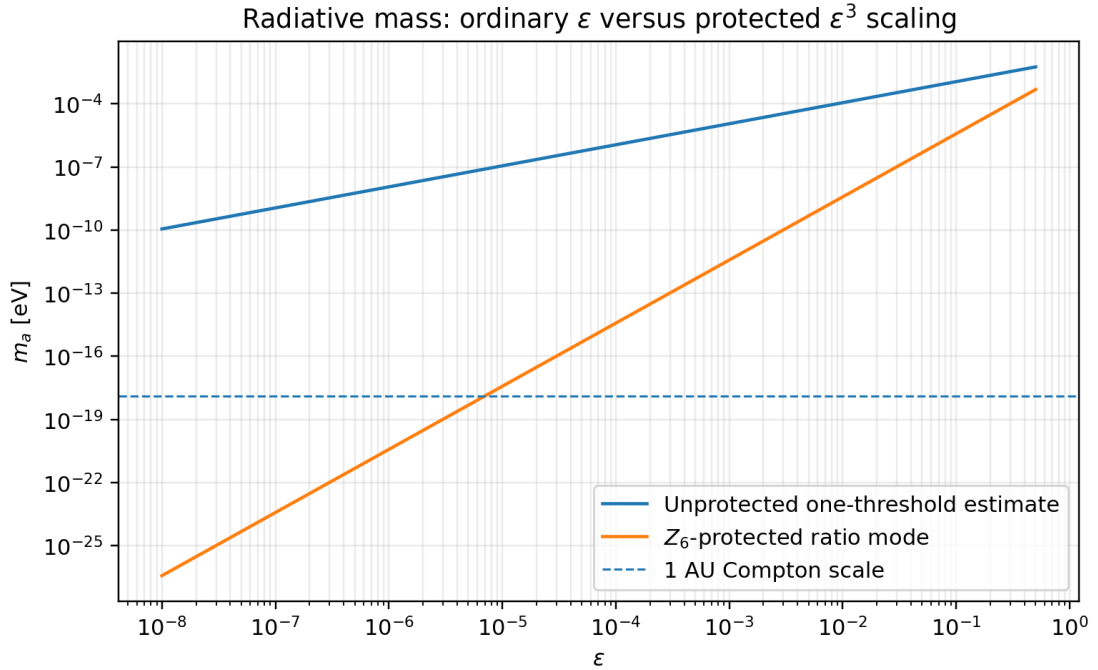


Figure 2: One-loop mass scaling for $M = 10 \text{ TeV}$ and $f_a = M_P$. The unprotected mode scales as $m_a \propto \epsilon$, whereas the Z_6 mode scales as $m_a \propto \epsilon^3$.

For the $N = 6, \epsilon = 10^{-6}$ point, the ordinary unprotected estimate is

$$m_{a,\text{naive}} \simeq 1.13 \times 10^{-8} \text{ eV},$$

whereas the protected result is

$$m_{a,Z_6} \simeq 3.80 \times 10^{-21} \text{ eV}.$$

The squared-mass suppression is

$$\frac{m_{a,Z_6}^2}{m_{a,\text{naive}}^2} = 1.125 \times 10^{-25}.$$

9 Chronometric and equivalence-principle predictions

The protection changes the naturalness relation; it does not erase the low-energy prediction derived in v0.5.

For clock transitions A and B ,

$$\mathcal{S}_{AB} \equiv \text{d} \ln \frac{\nu_A}{\nu_B}.$$

In the pure-QCD benchmark,

$$\mathcal{S}_{AB} = (\Delta K_\mu - \Delta K_q) \frac{2}{27} \kappa_0 \text{d}x + \dots$$

At the Z_6 benchmark vacuum, $\kappa_0 = \epsilon$, so

$$\mathcal{S}_{AB} = (\Delta K_\mu - \Delta K_q) \frac{2\epsilon}{27} \frac{\text{d}a}{f_a} + \dots$$

The rank-one relation remains

$$\mathcal{S}_{\text{Sr/Cs}} = 2\mathcal{S}_{\text{Sr/CaF}}$$

under the same leading sensitivity assumptions used in v0.5.

For one unscreened scalar and one dominant QCD spurion, the clock–equivalence-principle consistency line also remains

$$\frac{\beta_{AB}}{\eta_{CD}} = -\frac{\Delta K_\mu - \Delta K_q}{\Delta Q'_{\hat{m}}{}^{CD}}.$$

The Z_N sector changes the technically natural relation between m_a and the common coupling d_g ; the coupling cancels from the dimensionless clock/EP ratio.

10 Exact symmetry requirements and failure modes

10.1 The complete renormalization environment must be replicated

It is not enough to add $N - 1$ arbitrary spectator fermions. If Ψ_0 interacts with visible QCD while its partners see different gauge or matter content, gauge corrections split their masses and Yukawa functions. That splitting is a hard Z_N -breaking spurion and regenerates low harmonics.

The safe microscopic options are:

1. replicate the full Standard Model, as in existing Z_N -protected modulus constructions;
2. replicate at least every gauge and matter interaction entering the Ψ_k renormalization functions;
3. derive the exchange symmetry from a discrete gauge or geometric translation symmetry.

10.2 Quantitative tolerance to hard breaking

Let sector-dependent corrections be represented by weights $1 + \Delta_k$. Define their discrete Fourier components

$$\tilde{\Delta}_p = \frac{1}{N} \sum_{k=0}^{N-1} \Delta_k e^{-2\pi i p k / N}.$$

An order- p harmonic can then appear parametrically as

$$\delta V_p \sim \frac{N_c M^4}{16\pi^2} \tilde{\Delta}_p \epsilon^p \cos(px + \delta_p).$$

For the protected N -th harmonic to dominate,

$$|\tilde{\Delta}_p| \lesssim \epsilon^{N-p}, \quad p < N,$$

up to order-one coefficient ratios.

For $N = 6$ and $\epsilon = 10^{-6}$, this is ferocious. The symmetry must be exact in the Lagrangian; approximate sector equality is not enough.

10.3 Elementary-field quality problem

The operator

$$\frac{c_N}{M_*^{N-4}} \left(\Phi^N + \Phi^{\dagger N} \right)$$

is allowed by Z_N . It produces

$$\delta m_a^2 \sim N^2 |c_N| \frac{f_a^{N-2}}{M_*^{N-4}}.$$

For a high decay constant, this can overwhelm the protected fermion-loop potential. Therefore the elementary-global- Φ theory is an infrared model, not a sufficient ultraviolet explanation.

A credible completion must make the approximate continuous shift a gauge or geometric remnant, or otherwise suppress c_N by a controlled microscopic mechanism.

11 Preferred quality completion: a deconstructed Wilson-line ratio mode

The most attractive ultraviolet direction is to realize a as the gauge-invariant Wilson-line phase of a four-dimensional deconstructed gauge theory.

Take a cyclic moose of gauge groups connected by link scalars. Local gauge transformations remove all but one collective phase. The remaining gauge-invariant variable is schematically

$$\exp\left(i\frac{a}{f_a}\right) \sim \prod_j \frac{\Sigma_j}{|\Sigma_j|}.$$

Its shift symmetry descends from gauge invariance. Local renormalizable operators cannot depend on the full Wilson loop; an a -dependent potential requires propagation around the complete theory-space circle or explicit boundary/nonlocal breaking. This supplies:

- a discrete/geometric origin for the cyclic exchange;
- protection against generic local ultraviolet operators;
- a natural explanation for why the first field-dependent invariant requires a complete orbit;
- a route to a fully renormalizable four-dimensional quality completion.

A complete model must still derive the separate-QCD-sector mass orbit while keeping the colour groups distinct. The technically clean implementation is a replicated colour quiver linked only through a gauge-protected singlet Wilson-line sector and symmetry-related messenger chains. This is a concrete next construction rather than a solved detail of the present note.

12 Cosmological and environmental issues

12.1 Mirror sectors and reheating

Exact microscopic Z_N implies degenerate sectors. If all are reheated equally, hidden radiation and relics are generally unacceptable. The symmetry should be broken by the **state**, not by hard couplings:

exact symmetric Lagrangian + asymmetric initial condition/reheating state.

A reheating model must preserve the ultraviolet cancellation while populating primarily one sector. Recent cyclic models explicitly exploit spontaneous asymmetry of the relic state while retaining exact microscopic symmetry.

12.2 Finite-density potential

Visible matter is itself a sector-asymmetric state. It generates

$$V_{\text{matter}}(a) = \rho_{\text{vis}} \left[1 + \alpha_{\text{vis}} \frac{a - a_0}{M_{\text{P}}} + \cdots \right].$$

This term is not ϵ^N -suppressed. It is the physical source of the fifth force and clock redshift, but it can also:

- shift the local vacuum;
- increase the in-medium effective mass;
- screen or pin the field inside dense bodies;
- create nontrivial boundary profiles between sectors of different density.

The next phenomenological calculation must solve the nonlinear static field profile for Earth, Sun and laboratory source geometries.

12.3 Domain walls

The protected vacuum has N degenerate minima. Domain walls are avoided if:

- symmetry breaking occurs before inflation and is not thermally restored;
- a tiny symmetry-compatible bias removes the degeneracy without spoiling the mass hierarchy;
- or the discrete symmetry is gauged/geometric so apparently distinct vacua are identified appropriately.

12.4 Heavy coloured fermions

A stable visible colour triplet is excluded cosmologically. The visible Ψ_0 must decay. A vectorlike down-type or up-type assignment with small replicated mixing can provide decays while preserving the one-loop fundamental-representation threshold coefficient. The exact collider bound is decay-model dependent and is not fixed in this note.

12.5 CP structure

The mass orbit is real. Integrating out Ψ_k produces $aG_k^{\mu\nu}G_{\mu\nu}^k$ -type scalar couplings, not an anomalous $aG\tilde{G}$ term from a complex fermion mass phase. A generalized CP assignment and the chosen vacuum must nevertheless be specified in a full phenomenological model.

13 Novelty boundary

The broad protection mechanism is established:

- nonlinearly realized discrete symmetries can permit nonderivative Yukawa couplings while forcing the radiative potential to begin at high spurion order;
- replicated-Standard-Model Z_N models have used the same orbit cancellation to keep a photon-coupled modulus ultralight;
- mirror-QCD Z_N constructions suppress axion masses while preserving comparatively large couplings;
- a 2026 axion-WIMP model again uses cyclic fermions to project the Coleman-Weinberg potential onto the N -th harmonic.

Therefore the following cannot be claimed as new:

“use Z_N copies to obtain $V \sim \epsilon^N$ while a one-sector coupling is $O(\epsilon)$.”

The candidate original contribution is narrower:

1. a **CP-even vectorlike QCD threshold** rather than a photon threshold or anomalous axion coupling;
2. preservation of the exact visible-QCD decoupling response

$$\mathcal{D}_{\Psi \rightarrow 3}^{(1)} = \frac{2}{27};$$

3. embedding that response in the universal-scale-lock/chronometric-shear framework;
4. the Z_6 mass-coupling law

$$m_a^2 \propto d_g^6;$$

5. the observation that cyclic protection leaves the dimensionless clock-EP consistency line unchanged while making an ultralight mediator technically possible.

A systematic priority search found close precedents for every mechanism, but no exact indexed match for this combined QCD-threshold/chronometric construction. That remains a candidate priority claim, not a declaration of priority.

14 What has been solved and what has not

14.1 Solved in this step

$$\begin{aligned} \text{exact cyclic symmetry} &\implies V_{\text{vac}}(a) = O(\epsilon^N), \\ \text{one visible fundamental threshold} &\implies d \ln(\Lambda_3/\chi) = \frac{2}{27} d \ln(M_0/\chi) + \dots, \\ N = 6, \ x_0 = \pi/2 &\implies m_a^2 = \frac{27}{320\pi^2} \frac{M^4}{f_a^2} \epsilon^6, \quad d_g = \frac{2}{27} \frac{M_{\text{P}}}{f_a} \epsilon. \end{aligned}$$

This removes the catastrophic $M^4 \epsilon^2 / f_a^2$ contribution without nullifying the desired QCD response.

14.2 Still open

1. construct the full discrete-gauge/deconstructed messenger completion that produces the orbit while keeping $SU(3)_k$ sectors distinct;
 2. derive two-loop gauge/Yukawa corrections and verify the $O(\epsilon^N)$ spurion counting in the complete replicated theory;
 3. solve asymmetric reheating without hard Z_N breaking;
 4. calculate finite-density screening around realistic sources;
 5. specify the decaying vectorlike-quark phenomenology;
 6. combine current clock, MICROSCOPE, torsion-balance, stellar and cosmological limits in one likelihood;
 7. test whether the same protection can arise from the proposed deeper 36-field/Turok sector rather than an added elementary Φ .
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15 Immediate next calculation

The next decisive calculation is no longer the vacuum mass. It is the **environmental field profile**. For the Z₆ benchmark, derive and solve

$$\square a = \frac{\partial V_{Z_6}}{\partial a} + \sum_A \frac{\partial \ln m_A(a)}{\partial a} \rho_A(x)$$

for spherical sources, including the QCD composition charges of Earth, Sun and laboratory test bodies. This will determine whether the same finite-density effect that makes the model observable also screens it.

A parallel ultraviolet task is to construct the deconstructed gauge-quality completion and compute its leading nonlocal breaking operator. Those two tasks now dominate the viability question.

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17 Verification statement

The accompanying verification script independently checks:

- the closed form of F_N ;
- root-of-unity cancellation for $m < N$;
- the numerical one-loop Fourier coefficient for representative N ;
- the telescoping 2/27 threshold coefficient;
- the exact Z₆ identities $F_6 = 1/160$, $\kappa_0 = \epsilon$ and $m_a^2 = (27/320\pi^2)M^4\epsilon^6/f_a^2$;
- positivity of all M_k over 100,000 random points with $0 < \epsilon < 1$;
- representative mass/range benchmarks.

All programmed checks pass.